the target with a gas (usually He) containing aerosol particles (often KCl). The formed products have enough energy to recoil out of the target, if it is thin enough, and recoiling ionized atoms attach them selves to the aerosol particles. They can then be swept away several tenths of meters within a few seconds to a low-background area where the chemistry is performed.

When using a gas-filled mass separator between the target and the chemistry there is an additional problem because the products have to pass from the low pressure ($\sim$100 Pa) in the mass separator to the high pressure ($\sim$0.2 MPa) in the gas jet transportation system. This is achieved by using a recoil transfer chamber with a thin window that allows the nuclear reaction products to pass from the low pressure to the high pressure area where they attach to the aerosol particles.

17.5. PRODUCT SPECIFICATIONS

In many countries there are one or more suppliers of *radiochemicals*. To locate suppliers, the simplest way is often to browse the internet but it is also possible to contact the nearest nuclear center, as it may be a producer of radionuclides. Other sources of information may be e.g. the national radiation safety organization, or the annual Byers' Guide of common nuclear journals.

The radionuclides come in the form of calibration/reference sources, and as inorganic compounds and labelled organic compounds. They are packed in various ways, most commonly for compounds in a glass ampoule packed in a small aluminium can. To reduce the surface radiation dose rate (cf. Ch. 15) for high intensity radiation sources (usually $>$10 MBq) the can is put in a lead block which is packed in a wooden crate for shipment. Opening of the aluminum can and glass ampoule may require remote control and should be done only by experienced, licensed personnel.

17.5.1. Radiochemical processing

In some cases it may be a long process between the removal of the irradiated target from the irradiation facility and the pure radiochemical substance or compound to be used in some experiment. The separation procedures needed to make a pure radiation source are chemically the same and (usually) independent of the "size" (i.e. its total radioactivity), but the experiments become more cumbersome — and often slower — when the radiation source is so large that considerable shielding is needed to perform the work.

Most chemical techniques can be applied in the purification processing: precipitation, ion exchange, solvent extraction, electrodeposition, electrophoresis, distillation, etc. The basic purposes are to eliminate radioactive contaminants and to avoid diluting the radionuclide by isotopic stable atoms. If the radionuclide or radiochemical is to be used in medical and biological work, the sample may have to be sterile. Some publications about radiochemical separation procedures are given in the literature list, §17.9.

17.5.2. Specific activity

The specific activity or concentration of a radionuclide, i.e. radioactivity per gram, mole, or volume, is a very important factor in radionuclide work, e.g. in chemical work with substances of low solubility, in microbiology, etc.

Radionuclides of high specific activity are produced either through accelerator irradiation or through secondary reactions in the target (§17.6) in a reactor. Maximum specific activity is obtained when the radioactive nuclide is the only isotope of the element present in the sample. This is not possible to achieve in regular reactor irradiation through (n,γ) capture processes. For example, reactor-produced ^{24}Na may be obtained in specific activities of $\leq 2 \times 10^{11}$ Bq g^{-1}, while the specific activity of accelerator-produced ^{24}Na may exceed 10^{13} Bq g^{-1}; however, the total activities available are usually the inverse.

A *carrier-free* radioactive sample is usually one in which the radionuclide is not diluted with different isotopes of the same element. In reactor production of ^{24}Na from target ^{23}Na, each ^{24}Na is diluted with a large number of ^{23}Na atoms. ^{24}Na cannot be made carrier-free in a reactor. If a carrier-free radionuclide has been produced, e.g. through accelerator irradiation, which then must be purified, its concentration is so low that it may not follow the normal chemical rules. A macroscopic amount of *carrier*, either isotopic or not, may have to be added to carry the radionuclide through the proper chemical purification steps. We discuss this further in §18.2.

17.5.3. Labelling

A radioactively *labelled* (or "*tagged*") compound is a compound whereat least one of the atoms are radioactive. Preparation of labelled compounds may involve lengthy chemical synthesis starting with the radioactive nuclides in elementary form or in a simple compound.

Most syntheses of ^{14}C-labelled organic or biochemical compounds follow conventional methods with appropriate modifications to contain the radioactivity. $^{14}CO_2$ gas liberated by action of an acid on labelled $Ba^{14}CO_3$ is used commonly as a starting point in the syntheses. Three examples can illustrate how compounds can be labelled at different sites of the molecule:

1. Acetic acid-2-^{14}C:

$$\begin{aligned} Ba^{14}CO_3 &\xrightarrow{HCl} {}^{14}CO_2 \xrightarrow[\text{catalyst}]{H_2} {}^{14}CH_3OH \xrightarrow{PI_3} {}^{14}CH_3I \xrightarrow{KCN} \\ {}^{14}CH_3CN &\xrightarrow{H_2O} {}^{14}CH_3CO_2H \end{aligned} \tag{17.50}$$

2. Acetic acid-1-^{14}C:

$$Ba^{14}CO_3 \xrightarrow{HCl} {}^{14}CO_2 \xrightarrow{CH_3MgBr} CH_3{}^{14}CO_2H \tag{17.51}$$

3. Acetic acid-1,2-^{14}C:

$$\text{(a) } Ba^{14}CO_3 \xrightarrow[\text{heat}]{Mg} Ba^{14}C_2 \xrightarrow{H_2O} {}^{14}C_2H_2 \xrightarrow[\text{catalyst}]{H_2O} {}^{14}CH_3{}^{14}CHO \xrightarrow{[O]} {}^{14}CH_3\ {}^{14}CO_2H \tag{17.52}$$

$$\text{(b) } K^{14}CN \xrightarrow{{}^{14}CH_3I} {}^{14}CH_3{}^{14}CN \xrightarrow{H_2O} {}^{14}CH_3{}^{14}CO_2H \tag{17.53}$$

In these examples labelling is said to be *specific* in that only one specific position in the compound contains a radioactive nuclide. Alternatively, labelling may be *general* when most, or all, positions are labelled, which often is the case for tritium-labelled compounds (cf. §17.6.2).

Complicated organic compounds, e.g. penicillin, can be labelled through *biosynthesis*. For example, if penicillin is grown in a substrate containing a simple ^{35}S compound, ^{35}S is incorporated into the penicillin mold. Since other ^{35}S-containing products may be formed also, the ^{35}S penicillin must be purified, e.g. through solvent extraction or paper chromatography (§18.2.6).

Thousands of labelled organic compounds are commercially available and some organizations offer labelling service on request.

17.5.4. Radiochemical purity

An important consideration in the use of radionuclides is their radiochemical purity since, should several radionuclides of different elements be present in a tracer sample used in an experiment, the result could be ambiguous and misleading. In a radio-chemically pure sample, all radioactivity comes from a single radioactive element. If the radioactivity comes from a single isotope, the sample may be said to be *radio-isotopically pure*.

The radioisotopic purity of a sample may be ascertained by measuring the half-life. This method obviously can be used only with radionuclides that show sufficient decay during the time of observation for a reliable half-life to be calculated. For radioactivity of longer half-lives the radioisotopic purity can be checked by measurement of the type and energy of the emitted radiations (Ch. 9) or by processing the tracers through one or more chemical steps characteristic of the element of which the radioactivity is presumably an isotope. For example, in the use of ^{90}Sr ($t_{½}$ 28.5 y), the radioactivity can be processed

through a number of typical reactions characteristic of strontium. If the radioactivity follows the expected chemistry, then the experimenter can be rather certain that it is strontium. This chemical check does not preclude the possibility that there might be several isotopic nuclides in the sample. However, the presence of several isotopes need not be a handicap in tracer studies concerned with the characteristic properties of a particular element.

When a compound labelled with a radioactive nuclide is used to study molecular behavior, it is necessary to ascertain the *radiochemical purity of the compound* and not just the purity of the radioactive isotope. This means it is necessary to be certain that the radioactivity comes only from a specific element and only from a specific compound incorporating that element. This can be determined through selective chemical operations in which it is frequently desirable that the compound should not be destroyed. Examples of such methods involve preparative gas chromatography, thin-layer chromatography, paper chromatography, dialysis, ion exchange, and solvent extraction. Checking that an element or compound is radiochemically pure is important not only for sources of samples prepared in the laboratory but for commercial samples as well. For example, ^{14}C-tagged organic compounds obtained commercially may have been prepared at varying times prior to purchase. Frequently, the decomposition of molecules in the sample due to the radiolysis that accompanies the radioactive decay results in the presence of products that interfere with the course of the experiment for which the labelled compound is to be used.

In some cases the radioactive product after a period of storage will have grown a sufficient amount of radioactive daughter that it is no longer sufficiently pure radiochemically. In these cases, the daughter activity must be separated prior to use. For example, the ^{90}Y normally present in a ^{90}Sr sample must be eliminated if the tracer solution is to be used to study strontium chemistry.

17.6. RECOIL SEPARATIONS

17.6.1. Target recoil products

When a projectile of high energy reacts with a target atom, the reaction products are emitted in the direction of the projectile, measured in laboratory coordinates. We use the term *recoil* to describe this. The products, whose kinetic energy can be calculated by the conservation laws of Ch. 10, are slowed and eventually stopped through collision with other atoms. Consequently, if the target is relatively thin, there is a certain probability that the recoil range of the product exceeds the thickness of the target and the target atoms can therefore escape the target. In such cases the recoiling products be caught on special foils behind the target (*catcher foils*). These catcher foils can be stationary or they can be a moving band or metal or plastic, or even a gas, which sweeps past the target material.

The recoil technique was used in the discovery of the first transuranium element, neptunium, by E. McMillan and P. H. Abelson in 1939. A uranium salt, $(NH_4)_2U_2O_7$,